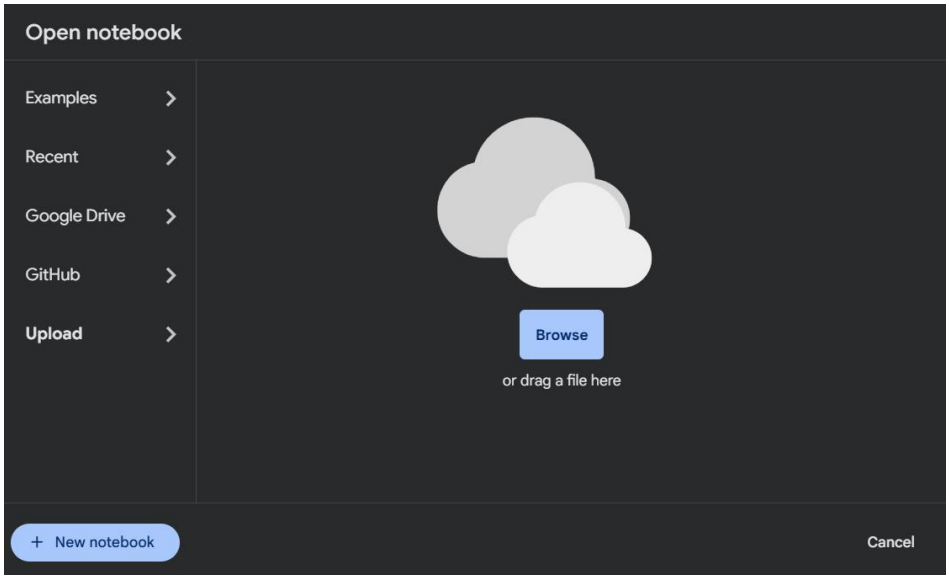
Getting started with Google Colab

Anyone with a Google account can run code in the web with minimal setup. It helps to have a folder prepared in your Google Drive to store all related materials, but it is not essential.

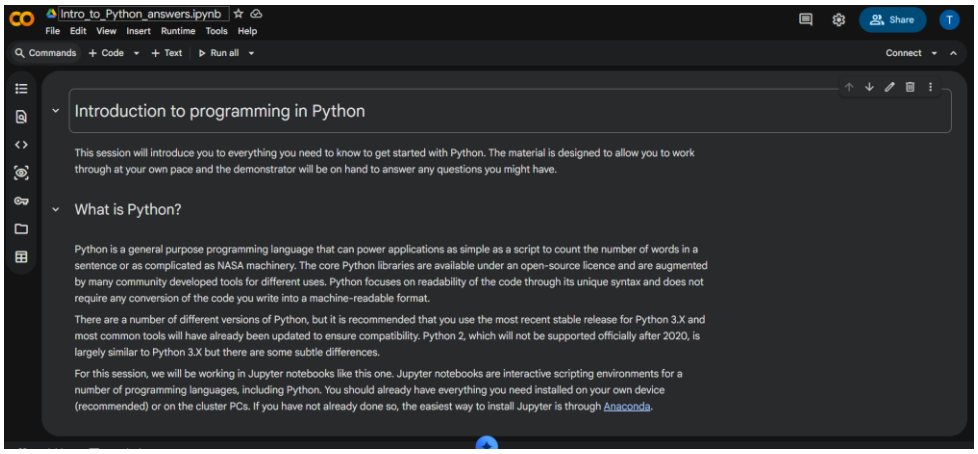
Step 1: Opening Colab

Opening Google Colab is as simple as navigating to https://colab.research.google.com/. Once you're signed in, you should see a pop up like the following:



It is asking you to provide a notebook, the format Colab expects code to be written in. This is the same as a Jupyter notebook if you've come across it through other projects.

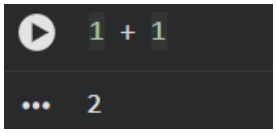
You can load a file from a space in your Google Drive, or just upload one you stored on your computer. Once you've selected the appropriate option, you should see something like the following:



At this stage you're ready to begin coding!

Step 2: Running code

The notebook is structured into cells, sections of text or code that are managed individually. Code cells can be executed within Colab and can be identified by the different font and the “run” button:



Running a code cell can be done in one of two main ways: clicking the run button, or pressing Shift+Enter when the cell is selected. At this point, Colab will run any code written within the cell and return output as appropriate for the commands given.

If you code executes successfully, you should see something like

The number in square brackets indicates the order in which the cells have been run, and the time next to the tick tracks how long it took for the cell to evaluate. If you don’t see the green tick, there has likely been an error with your code – check the cell output to see what went wrong.

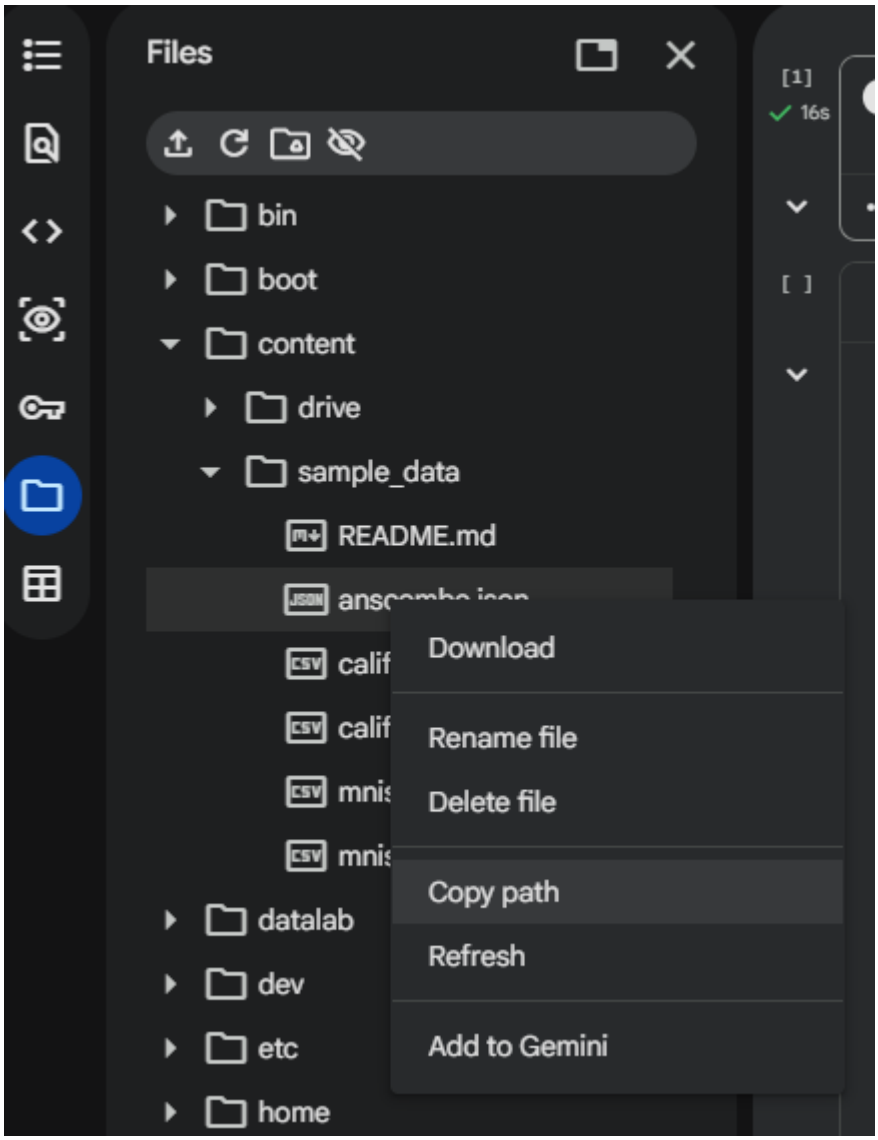
Step 3: Managing your environment

Colab requires a runtime to do the code execution. Luckily, it will manage everything for you. The first time you run a cell, you will be allocated some resources in Google Cloud to support your experiments. This is usually enough for most projects, and can even include the powerful hardware needed to train your own machine learning models.

The provided runtimes also include many of the standard libraries used within Python – this presents one of the major advantages with working in Colab rather than an installation on your own machine.

Step 4: Mounting your drive

Occasionally you will need to work with external files, e.g. for data or to save figures for later use. To do this in Colab you will need to "mount" your Google Drive. This is as simple as including a short snippet of code at the start of your notebook:

```
from google.colab import drive
drive.mount('/content/drive')
```

You can then reference any file in your Google Drive, but be careful to include the full path (i.e. the folders needed to reach it). You can always check this on the fly using the file browser on the left toolbar, and copy the required path.